Gopi Krishna

ce24btech11023@iith.ac.in | gopikrishna9968@gmail.com | +91 6900529119

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	6.36
GREEN VALLEY INT PUB SCHOOL	Class XII (CBSE)	2024	84.80%
KENDRIYA VIDYALAYA NHPC DHEMAJI ASSAM	Class X (CBSE)	2022	90.80%

SKILLS

- **Languages**: C, C++, Python
- **Concepts :** Data Structures & Algorithms
- **Dev Tools**: Git, GitHub

PROJECTS

URL Shortener | Next.js, TypeScript, Node.js, Express, PostgreSQL, Redis

- Built a full-stack URL shortening platform supporting **guest and authenticated users**, configurable link expiry, click tracking, and per-link analytics.
- Implemented a **Redis cache-aside architecture with a 24-hour TTL**, reducing repeated PostgreSQL queries and enabling low-latency redirects.
- Developed secure authentication using **JWT and bcrypt**, along with API rate limiting and user-specific link management.
- Designed an analytics dashboard using **Next.js and Recharts** to visualize daily click activity.
- Load-tested the redirect API using Autocannon with **100 concurrent connections**, achieving **~1,496 requests/second** and **~66 ms average latency** with a warm Redis cache.

Collaborative Project Management | Next.js, TypeScript, PostgreSQL, Prisma, Socket.IO, Redis

- Built a multi-tenant project management platform with **10,000+ seeded tasks** to simulate production-scale workloads.
- Added **composite indexes** and asynchronous background writes, reducing critical-path query latency to **~3 ms**.
- Improved drag-and-drop Kanban reordering from **O(n) to O(1)** using fractional indexing, reducing updates to a single-row write.
- Built real-time collaboration using **Socket.IO and Redis Pub/Sub** for low-latency multi-client synchronization.

Nash Shell -- Unix-like Command Shell

- Co-developed a modular Unix-like shell supporting external commands, built-in utilities, multi-stage pipelines, background processes, and I/O redirection (<, >, >>).
- Implemented process management using fork, execvp, waitpid, pipe, and dup2, including background-job tracking and child-process cleanup.
- Built a custom terminal interface using termios, featuring editable input, arrow-key navigation, command history, custom prompts, and non-canonical input handling.
- Added signal handling with sigaction and organized the codebase into separate parser, executor, terminal, history, built-in command, and job-management modules.
- Automated compilation and cleanup through a C++20 Makefile-based build system.

Physics Simulator -- C++20, Raylib

- Co-developed a real-time 2D physics simulator configured to model and render 100 moving bodies at a target of 120 FPS using C++ and Raylib.
- Implemented kinematic motion under gravity by updating object positions and velocities over fixed simulation time steps.
- Developed circle-to-circle collision detection and response using normal/tangent vector decomposition, overlap

correction, and a configurable coefficient of restitution.
- Added collision handling against window boundaries and randomized initial positions and velocities for dynamic simulations.
- Structured the application into reusable Body, Scene, Window, and Vec2 components, separating rendering, physics, collision handling, and state initialization.

RELEVANT COURSES

- **Math and CS courses**: Introduction to Programming, Digital Fabrication, Probability & Random Variables, Statistics, Linear Algebra, Calculus, Complex Variables, Transform Techniques
- **Physics:** Engineering Mechanics, Fluid Mechanics, Hydraulic Engineering, Thermodynamics

ACHIEVEMENTS

- Codeforces -- maximum rating: **1432**
- Ranked **769th out of 21,000+ participants** in Codeforces Round 1090 (Div. 4)
- Ranked **163rd** in CodeChef Starters 219